



1.4.5 Wrap-Up: Cool Down

1. Isometry is a transformation that preserves distance and angle measure. Write a short note that describes whether a dilation preserves distance or not.



Unit 1, Lesson 5: Transformations on the Plane – Part 1



Benchmark

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.



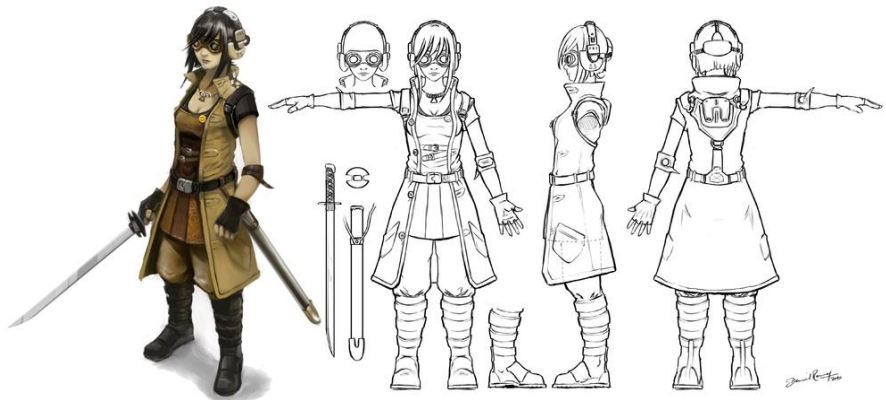
Learning Targets

- ☐ I can identify the angle of rotation of a transformation of a figure on a plane.
- ☐ I can identify the center of rotation of a transformation of a figure on a plane.
- ☐ I can identify the direction of a rotation of a transformation of a figure on a plane.
- ☐ I can distinguish a rotation from other transformations.



1.5.1 Warm-Up: Rotations in Drawings

The picture shows the front, side, and back view of an animated character.

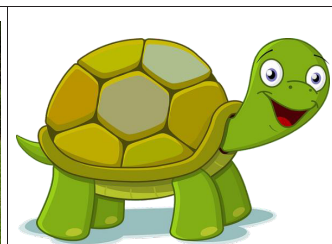
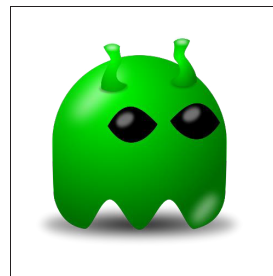


1. What do you notice about all three views — front, side, and back, of the animated character?

When animators and cartoonists sketch the movement of a character, they use a sheet like the one shown.

Front	Side	Back

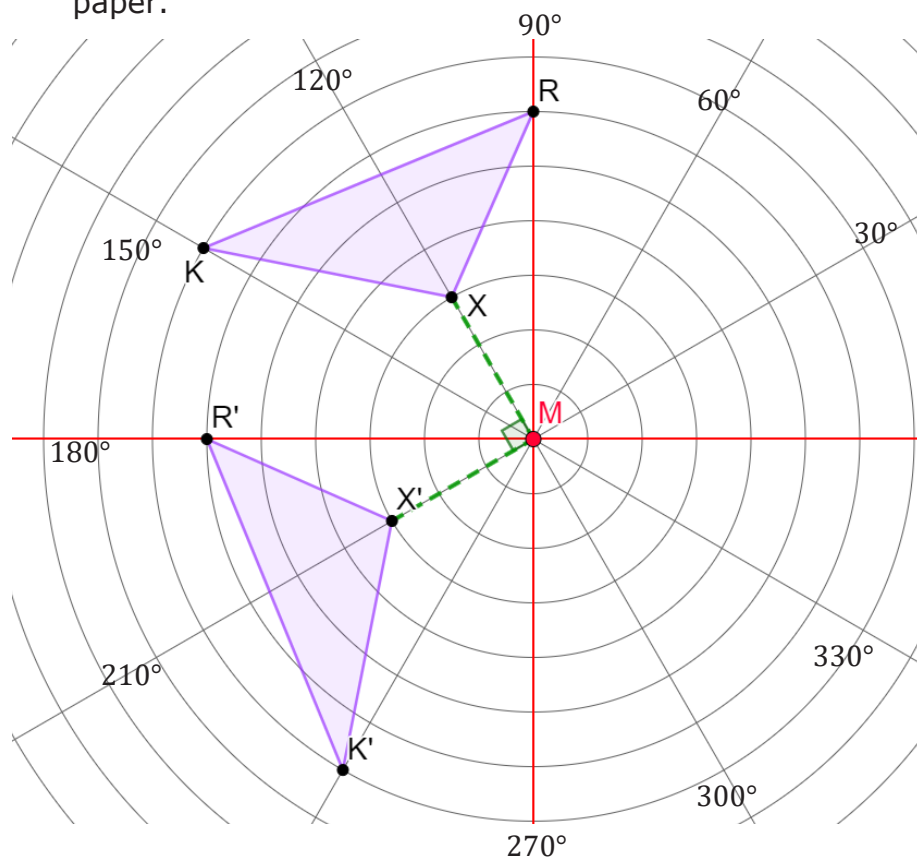
2. Use one of the pictures below, or your own idea, to sketch the three views. Use the dotted lines to help you align key features of the picture.





1.5.2 Guided Instruction: Transformations on the Plane

1. The diagram shows $\triangle RKX$ and $\triangle R'K'X'$ on circular grid paper.



Follow the steps below.

- Use a piece of patty paper to copy $\triangle RKX$ and point M .
- Place the patty paper so that it aligns with $\triangle RKX$ and point M .
- Place the point of your writing utensil on point M and rotate $\triangle RKX$ so that it lays on top of $\triangle R'K'X'$.

- a. Draw $\angle RMR'$.

- b. Draw $\angle KMK'$.

- c. Complete the table.

Angle	Measurement (degrees)
$\angle XMX'$	
$\angle RMR'$	
$\angle KMK'$	

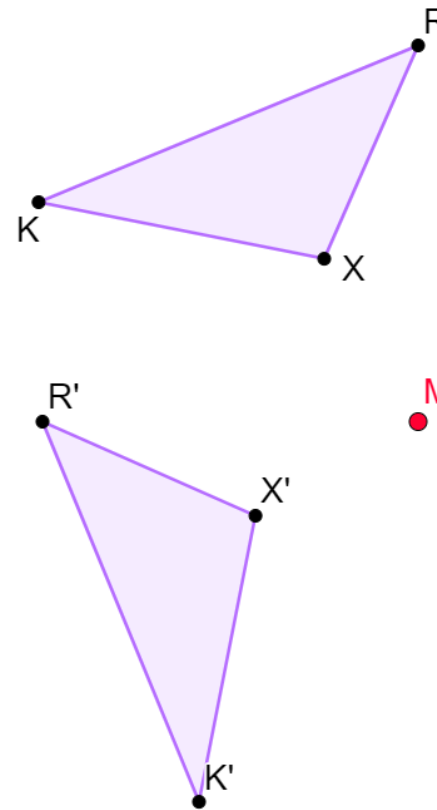
- d. Discuss with your partner how the angle measures on the circular grid can be used to verify the rotation and how to determine the direction of rotation. Summarize your discussion.

- e. Identify the transformation of $\triangle RKX$. Describe the transformation using direction, degrees, or scale factor.

Type of Transformation	Center, Line, or Movement	Description
○ Dilation ○ Rotation ○ Reflection ○ Translation	○ Center of dilation is _____. ○ Center of rotation is _____. ○ Line of reflection is _____. ○ Down ○ Left ○ Right ○ Up	

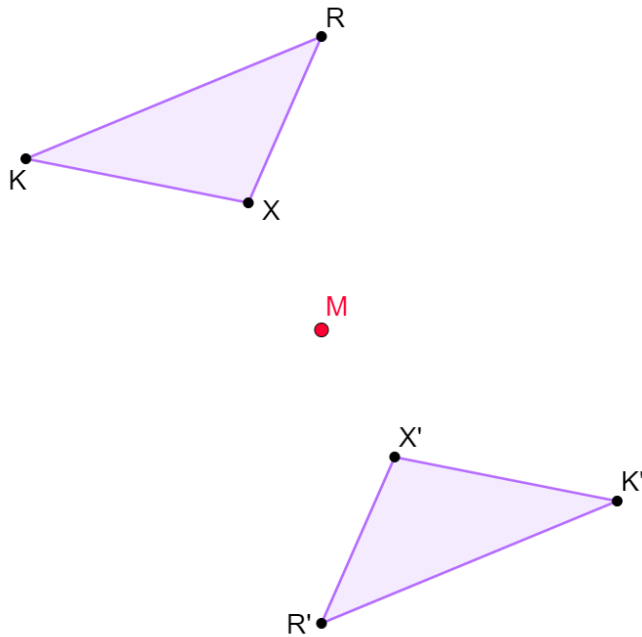
- f. Discuss with your partner whether or not the transformation could be described as “a rotation of 270° clockwise centered at point M .” Explain why or why not.

2. The diagram shows $\triangle RKX$ and $\triangle R'K'X'$ without the circular grid paper.
- a. Draw $\angle XMX'$, $\angle RMR'$, and $\angle KMK'$.



- b. Discuss with your partner how the degree of rotation can be determined using the angles that connect the vertices and the center of rotation.

3. The diagram shows $\triangle RKX$ and $\triangle R'K'X'$ using a different transformation.



- a. Identify the transformation of $\triangle RKX$. Describe the transformation using direction, degrees, or scale factor.

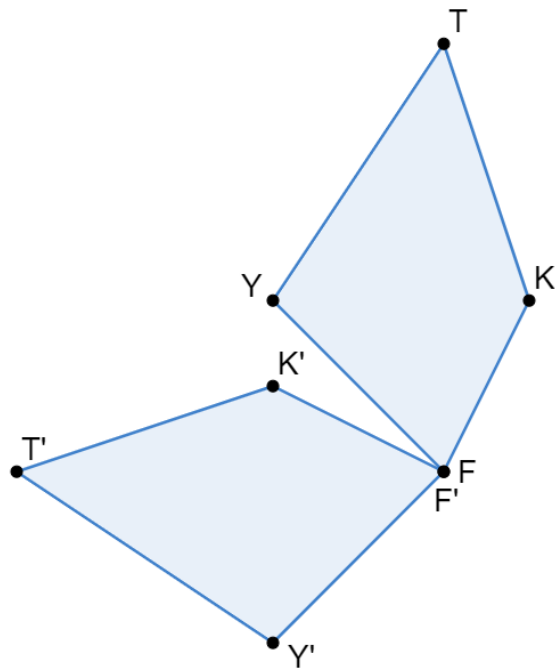
Type of Transformation	Center, Line, or Movement	Description
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	

- b. Two students' descriptions of the transformation of $\triangle RKX$ are shown.

Michela	Marc
a reflection across line M	a translation down

- Use a piece of patty paper to copy $\triangle RKX$.
 - Perform each student's transformation to prove that each transformation does not work.
- c. Discuss with your partner why both students are wrong. Write your reasoning for why the transformations do not work.

d. The diagram shows $TKFY$ and $T'K'F'Y'$.



Identify the transformation of $TKFY$. Describe the transformation using direction, degrees, or scale factor.

Type of Transformation	Center, Line, or Movement	Description
○ Dilation ○ Rotation ○ Reflection ○ Translation	○ Center of dilation is _____. ○ Center of rotation is _____. ○ Line of reflection is _____. ○ Down ○ Left ○ Right ○ Up	



1.5.3 Guided Practice: Rotations

The table shows three different rotations of $TKFY$. Determine the direction of rotation, center of rotation, and the degree of rotation for each.

Transformation	Direction, Center, and Degree
	• Direction of rotation is _____. • Center of rotation is _____. • Degree of rotation is _____.
	• Direction of rotation is _____. • Center of rotation is _____. • Degree of rotation is _____.



- Direction of rotation is _____.
- Center of rotation is _____.
- Degree of rotation is _____.

1.5.4 Wrap-Up: Ticket Out the Door

1. The table shows two different rotations of $TKFY$. Determine the direction of rotation, center of rotation, and the degree of rotation for each.

Transformation	Direction, Center, and Degree
	• Direction of rotation is _____. • Center of rotation is _____. • Degree of rotation is _____.
	• Direction of rotation is _____. • Center of rotation is _____. • Degree of rotation is _____.